An analysis of performance of different pose estimation toolboxes

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***Abstract*—As part of our research project we aim to evaluate which method is best suited for pose estimation on toads and pose tracking on shrimps.**

***Index Terms*—pose estimation, pose tracking, performance analysis, toads, shrimps**

I. Introduction

Animal pose estimation (APE) aims to locate the animal body parts using a diverse array of sensor and modality inputs (e.g. RGB cameras, LiDAR, infrared, IMU, acoustic and language cues).

This paper is conducted within the framework of the CS8813 GTL course Introduction to Research and serves as a comprehensive evaluation of contemporary 2D animal pose estimation techniques applied to real-world, non-standard biological datasets.

This work investigates the performance of state-of-the-art 2D RGB-based animal pose estimation methods on two biologically distinct and visually challenging datasets: amphibians (primarily toads) and aquatic invertebrates (shrimps). These species present unique difficulties, including non-human body morphology, frequent self-occlusion, small articulated structures, and low-contrast imaging conditions. In particular, the amphibian dataset consists of grayscale images captured using an infrared motion-detection camera, while the shrimp dataset comprises RGB video sequences requiring both pose estimation and temporal tracking.

The objective of this study is to systematically evaluate and compare existing 2D monocular pose estimation and tracking methods under a controlled experimental protocol. Each model is trained independently on identical datasets, one composed of static amphibian images and the other of shrimp video sequences, to ensure a fair comparison. Model performance is quantitatively assessed using standard pose estimation accuracy metrics, allowing for an objective analysis of their robustness and suitability for each species.

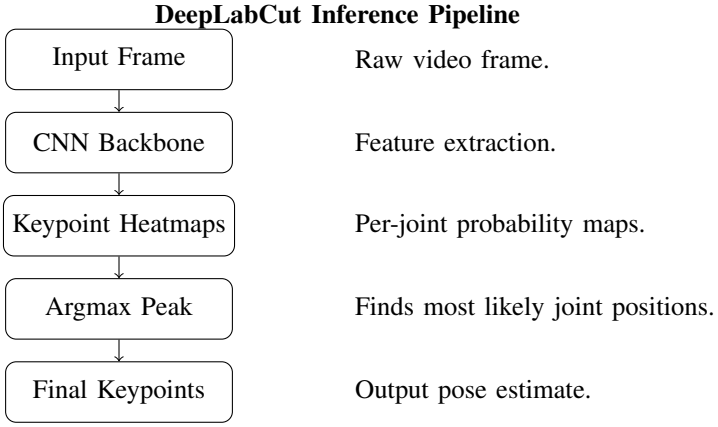
II. State of the Art

A. DeepLabCut: Markerless Feature Extraction

DeepLabCut (DLC) is a premier framework for 2D animal pose estimation that leverages deep convolutional neural networks, typically utilizing ResNet or EfficientNet backbones pretrained on ImageNet [7]. A significant advantage of DLC is its transfer learning capability; pretraining on large-scale datasets has been shown to significantly boost out-of-domain robustness, particularly when training data is scarce [6]. This allows the model to learn pose-specific features, such as keypoint relationships and skeletal constraints, enabling faster convergence even with limited task-specific annotations.

Despite its success, DLC has notable constraints in non-standard environments. As it primarily processes frames independently, it lacks inherent temporal context, making it sensitive to jitter and keypoint ”swapping” during occlusions. Furthermore, its performance often degrades when the target species (e.g., toads) differs significantly in morphology from the mammals represented in the Model Zoo, or when imaging conditions—such as the grayscale infrared footage used in this study—diverge from the standard training distribution.

DeepLabCut Inference Pipeline



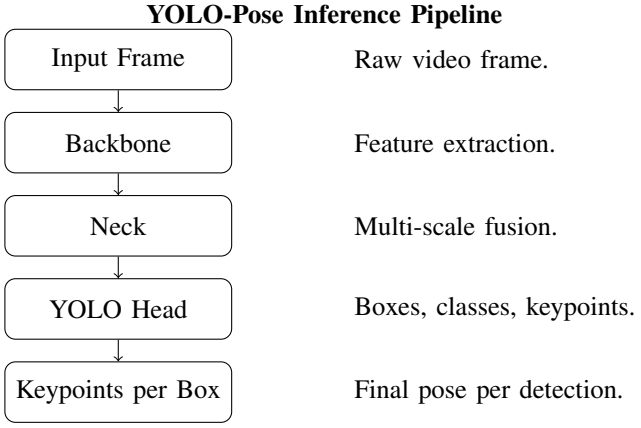
B. YOLO-based Pose and Posture Estimation

The YOLO (You Only Look Once) framework, specifically variants like YOLO-Pose, treats pose estimation as a single-stage regression problem, predicting bounding boxes and anatomical keypoints simultaneously [1], [2]. In aquatic monitoring, YOLO variants (v4, v7, and v8) have proven robust for detecting and counting small objects in cluttered, low-contrast environments [3], [4]. Recent advancements in YOLOv8 have further improved tracking accuracy by incorporating reflection and parallax correction, which is vital for underwater imaging [5].

For shrimp analysis, YOLO-Pose allows for the reconstruction of posture through geometric relationships between the head, thorax, and tail. This is particularly valuable for

non-destructive quality assessment, such as determining the freshness of a specimen based on body curvature. However, YOLO lacks explicit kinematic priors, which can lead to anatomically inconsistent predictions in highly flexible organisms. Its reliance on large, high-quality annotated datasets also presents a challenge for niche biological species where public data is scarce.

YOLO-Pose Inference Pipeline



C. SLEAP: Multi-Animal Tracking and Identity

SLEAP (Social LEAP Estimates Animal Poses) is a framework explicitly designed for multi-animal scenarios where identity preservation is critical [10]. It supports two distinct grouping structures: *bottom-up*, which detects all body parts and assembles them into individuals, and *top-down*, which first detects individuals and then estimates pose within cropped regions. SLEAP is distinguished by its integrated identity tracking, which combines motion continuity with appearance-based models to maintain consistent labels across time.

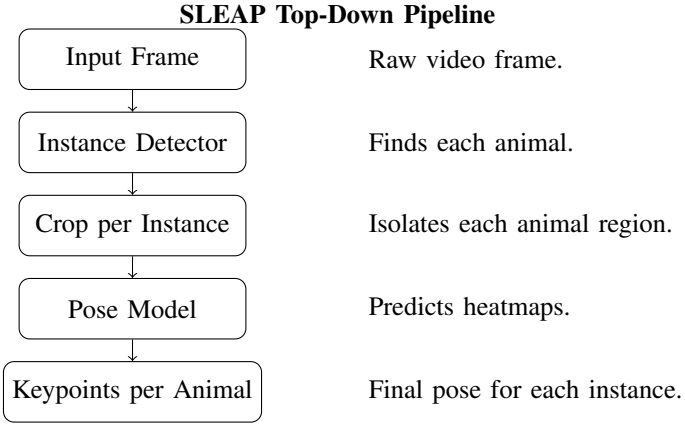
The versatility of SLEAP has seen it adapted for long-term deep phenotyping in home-cage monitoring [11] and even for tracking non-animal morphologies like plant growth dynamics, where it has proven robust despite changing 2D leaf shapes and occlusions [9]. Empirical evidence suggests that SLEAP is highly data-efficient, often achieving stable results with only 100–200 annotated frames, and its lightweight UNet-based architecture makes it a high-speed alternative for real-time video analysis [10].

D. Synthesis and Tool Selection

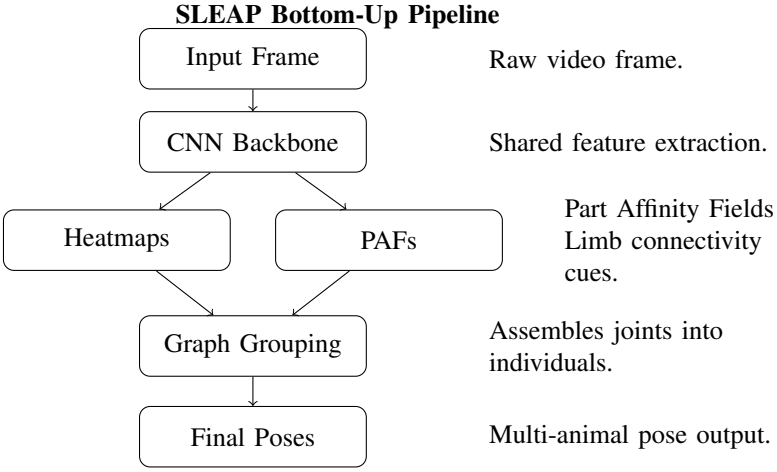
Each framework presents a distinct trade-off between architectural complexity and functional utility. For the amphibian dataset, which consists of static, low-contrast infrared images, DeepLabCut is hypothesized to be the most promising due to the established benefits of pretraining for out-of-domain robustness [6].

Conversely, for the shrimp dataset, SLEAP stands out as the superior candidate. Its native ability to handle temporal tracking and multi-animal grouping addresses the specific challenges of aquatic video sequences, specifically, maintaining identity through the fluid, overlapping movements of multiple subjects [10]. While YOLO-Pose offers high speed, the biological requirement for consistent temporal tracking suggests that SLEAP's integrated grouping and appearance-based identity models will yield more reliable behavioral data.

SLEAP Top-Down Pipeline



SLEAP Bottom-Up Pipeline



III. Methodology

A. Objective

The objective of this study is to benchmark three pose–estimation approaches— SLEAP, DeepLabCut, and a YOLO-based keypoint detector—on a standardized, pre-labeled dataset (Fly32). By evaluating their accuracy, robustness, and computational efficiency under identical conditions, we aim to identify which method is most suitable for subsequent application to our target datasets containing toads and shrimps. The Fly32 dataset serves as a controlled environment for fair comparison, while the resulting performance trends will inform expectations for generalization to new species and recording conditions.

B. Dataset

We use the Fly32 dataset provided in the official SLEAP documentation, which includes videos of freely moving *Drosophila* along with high-quality ground-truth keypoint annotations. These annotations are used as the reference for quantitative evaluation. The dataset is divided into training, validation, and test subsets to ensure reproducible benchmarking across methods.

C. Methods Compared

SLEAP is evaluated using its standard bottom-up architecture for single or multi-instance pose estimation. **DeepLabCut** is tested using a MobileNet-V2_1.0 backbone trained for heatmap regression. The **YOLO-based** approach (e.g., YOLOv8-Pose) employs a single-stage, anchor-free architecture. It simultaneously predicts object bounding boxes and the coordinates of K associated keypoints, offering a highly efficient, coupled detection and pose estimation pipeline compared to traditional top-down methods.

TABLE I
COMPARISON OF POSE ESTIMATION METHODS

Property	SLEAP	DLC	YOLO
Arch.	BU/TD	Heatmap	Keypoint
Train	Multi-inst.	Single/Multi	Det.+KP
Strength	Tracking	Accuracy	Speed
Weakness	Complex	Slow train	Lower prec.
Output	KP+IDs	KP	KP

D. Experimental Protocol

All three methods are trained on the same training split of Fly32 using identical data augmentations (rotation, scaling, and brightness jitter). Hyperparameters such as batch size, learning rate, and number of epochs are kept consistent whenever possible to ensure fairness. Inference is performed on the held-out test set, and runtime is measured on the same hardware configuration.

E. Evaluation Metrics

Performance is quantified using the following metrics:

- **RMSE** between predicted and ground-truth keypoints, The RMSE for a set of N keypoints is defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\mathbf{k}}_i - \mathbf{k}_i\right\|^2}, \tag{1}$$

  where $\hat{\mathbf{k}}_i$ and $\mathbf{k}_i$ denote the predicted and ground-truth coordinates of keypoint i, respectively. This metric captures the average localization error in pixel space.
- **PCK** (Percentage of Correct Keypoints) at multiple thresholds.
  The Percentage of Correct Keypoints (PCK) is computed as:

$$\mathrm{PCK}(\tau) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\left\|\hat{\mathbf{k}}_i - \mathbf{k}_i\right\| < \tau\right), \tag{2}$$

  where τ is a distance threshold in pixels, and $\mathbb{I}(\cdot)$ is the indicator function that equals 1 when the predicted keypoint lies within τ of the ground truth and 0 otherwise. PCK measures the proportion of keypoints that are localized accurately within a specified tolerance.
- **OKS** (Object Keypoint Similarity) and **mAP** (mean Average Precision).
  To standardize the evaluation of the YOLO-based method alongside the others, we use OKS, defined as:

$$\mathrm{OKS} = \frac{\sum_i \exp\left(-d_i^2/2s^2\kappa_i^2\right)\delta(v_i > 0)}{\sum_i \delta(v_i > 0)}, \tag{3}$$

  where d_i is the Euclidean distance between the predicted and ground-truth keypoint, s is the object scale, κ_i is a per-keypoint constant that controls falloff, and v_i is the visibility flag. Model robustness is then reported using the mAP across various OKS thresholds (e.g., mAP@0.5:0.95).
- **Inference speed** (frames per second),
- **Training time** and computational cost.

F. Generalization to Target Datasets

Although Fly32 differs morphologically from our target species (toads and shrimps), benchmarking on a controlled dataset allows us to compare the baseline relative strengths of each method. Generalizing to shrimps introduces specific morphological challenges, such as partial transparency, complex jointed appendages (e.g., pleopods, pereiopods), and a rigid exoskeleton, which require a custom-defined keypoint skeleton. We hypothesize that methods demonstrating higher accuracy and stability on Fly32 will also perform better on our target datasets, despite this domain shift. The selected method will later be fine-tuned and validated on a manually labeled subset of toad and shrimp frames.

IV. RESULTS

A. Quantitative Performance and Generational Comparison

To establish a comprehensive baseline for our pose estimation pipeline, all models were initially trained and evaluated on the official SLEAP Fly dataset. This dataset comprises approximately 1,500 annotated frames featuring a highly complex, 32-node skeletal structure. Benchmarking on this dense keypoint configuration provides a rigorous stress test for spatial accuracy and model capacity before transitioning to our target amphibian and shrimp datasets.

We evaluated YOLO architectures across different capacity scales (nano and extra-large) and generations (v8 and v26), comparing them against DeepLabCut (DLC) and SLEAP under identical hardware constraints. Quantitative metrics, including training time and inference speed, are summarized in Table II.

TABLE II
CROSS-FRAMEWORK PERFORMANCE COMPARISON (FLY DATASET: 1,500 FRAMES, 32 NODES)

Framework (Model)	RMSE (px) ↓	Speed (FPS) ↑	Train (h) ↓
YOLO (*yolov8n*)	**3.91**	**17.2**	**0.48**
YOLO (*yolo26n*)	4.23	13.9	0.76
YOLO (*yolo26x*)	3.52	1.4	4.78
DLC (*ResNet-50*)	5.85*	8.5	∼3.00
SLEAP (*UNet*)	3.80	**45.0**	∼3.00

*Mean inflated by extreme outliers (no bounding box constraint).

An intra-architectural comparison at the nano scale reveals an unexpected generational regression on this specific dataset. The older *yolov8n* architecture outperformed the newer *yolo26n* across all metrics: it achieved a lower localization error (3.91 px vs. 4.23 px), faster inference (17.2 FPS vs. 13.9 FPS), and quicker convergence (0.48 h vs. 0.76 h). This suggests that the internal architectural modifications introduced in the newer YOLO generation introduce unnecessary computational overhead and inductive biases that do not translate well to sparse, monochromatic biological domains like the Fly dataset. Consequently, scaling up to the heavy *yolo26x* is necessary to surpass the baseline accuracy of *yolov8n*, albeit at a severe cost to inference speed.

Furthermore, compared to standard frameworks, the lightweight *yolov8n* converged remarkably fast (0.48 hours) despite the complexity of predicting 32 distinct nodes. In contrast, DeepLabCut and SLEAP required approximately 3 hours to complete their training routines on the same 1,500 frames. SLEAP, however, demonstrated the highest raw inference speed (45 FPS), benefiting from its highly optimized bottom-up tracking architecture.

B. Spatial Accuracy and Error Distribution

To move beyond simple scalar metrics, we analyzed the spatial robustness of each model using the Percentage of Correct Keypoints (PCK) and the error distribution across the test set.

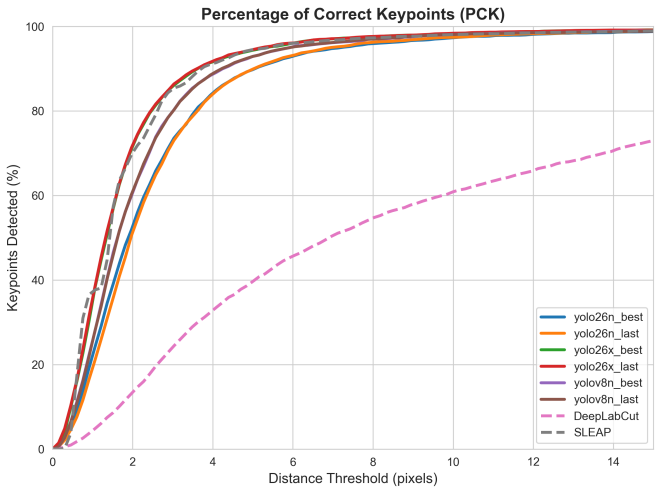


Fig. 1. Percentage of Correct Keypoints (PCK). The curve illustrates the proportion of accurately predicted keypoints across increasing pixel distance thresholds.

The PCK curve (Fig. 1) illustrates the models' reliability at various strictness thresholds. The YOLO variants and SLEAP demonstrate strong robustness, rapidly reaching high accuracy within a 10-pixel tolerance, which is notable given the high density of the 32-node skeleton. However, analyzing the global distribution of these errors reveals a critical architectural difference between the frameworks.

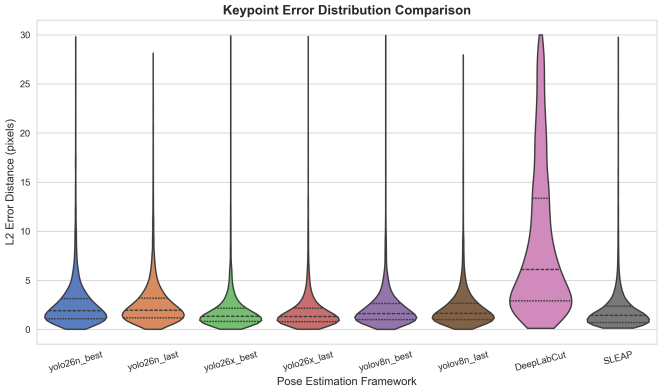


Fig. 2. Keypoint Error Distribution (L_2 distance). The extreme upper tail of the DeepLabCut distribution highlights severe outlier predictions caused by the absence of a bounding box constraint.

As shown in the Violin Plot (Fig. 2), DeepLabCut exhibits a highly concentrated core of pixel-perfect predictions, but suffers from extreme outliers, creating a long, extended tail in the error distribution. This phenomenon is a direct consequence of its architectural design. Unlike YOLO, which is a top-down approach that first detects a bounding box and constrains all 32 keypoint predictions within that localized area, DeepLabCut relies on a bounding-box-free, pure heatmap regression.

When DLC encounters ambiguous features or severe occlusions among the closely clustered nodes, it occasionally locks onto background noise, generating "ghost" predictions placed at the opposite end of the image. Because it lacks a bounding box to geometrically restrict the maximum possible error, these failures result in massive L_2 pixel deviations that severely penalize its overall RMSE. YOLO, by contrast, restricts its worst-case errors to the dimensions of the detected bounding box, resulting in a much tighter and more predictable error distribution.

C. Synthesis for Target Datasets

These comparative results directly inform our strategy for the amphibian and shrimp datasets. For the shrimp tracking task, which requires rapid processing of long video sequences, the older but highly efficient *yolov8n* or SLEAP are the optimal choices due to their high FPS and robust PCK scores. Conversely, for static, low-contrast amphibian images where real-time tracking is irrelevant, the bounding box mechanism of YOLO provides a crucial safeguard against the extreme outlier predictions observed in unconstrained heatmap approaches like DeepLabCut, especially when transitioning to complex skeletal structures.

V. Future Work

Future work will focus on extending the present evaluation beyond the Fly32 dataset while remaining mindful of the practical constraints associated with each target domain. Although the toad dataset was initially considered, we will not pursue experiments on this domain due to its strong discontinuity

relative to the training distribution and the substantial manual labeling effort required to establish a reliable ground truth. The combination of sparse motion, low frame-to-frame consistency, and the need to annotate a large number of images by hand makes this dataset unsuitable for the scope of the current project.

In contrast, the shrimp dataset presents a more promising direction. Its continuous motion and stable appearance across frames align well with SLEAP's strengths in multi-frame tracking and bottom-up pose assembly. Moreover, a preliminary ground truth set has already been generated through a rapid SLEAP inference pass, approximately 40 labeled frames, followed by light manual correction of outliers. This resulted in approximately 10000 corrected frames, providing a usable reference set for evaluating cross-domain generalization.

Building on these foundations, future work will involve retraining or fine-tuning SLEAP, DLC and YOLO-based models on the shrimp domain, assessing their robustness under low-contrast underwater conditions, and exploring whether the performance trends observed on Fly32 persist. Additional directions include incorporating scale-normalized metrics and keypoint importance such as OKS, evaluating temporal consistency across longer sequences, and investigating lightweight model variants suitable for real-time deployment.

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